

A Stastical Analysis of Equine Corneal Anesthesia

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Background and Motivation

- Corneal procedures in horses, like fixing corneal flaps or removing foreign bodies, are often done with the horse standing rather than under full anesthesia. This is because **general anesthesia carries about a 1% risk of serious complications during recovery.**
- To keep the horse comfortable and safe, we need a reliable way to **numb the surface of the eye locally**, so they don't feel pain but can still stand on their own.
- We routinely inject **ropivacaine** under the tissue around the eye to block sensation for several hours, but many clinicians add **dexmedetomidine** (a mild sedative) in the hope that it will extend that numbness even longer.
- It isn't yet proven whether adding dexmedetomidine really makes the cornea stay numb longer – or whether it changes other safety factors, like causing more bleeding where we inject.

Background and Motivation

- To find out, a controlled crossover study was ran, measuring four things:
 - ① **How long the eye stayed numb** (Total Anesthesia Time, or TTA)
 - ② **How deep the numbness went** (minimum corneal touch threshold)
 - ③ **How long the deepest numbness lasted** (duration of minimum CTT)
 - ④ **Any bleeding at the injection site** (hemorrhage score)

What is Corneal Touch Threshold (CTT)?

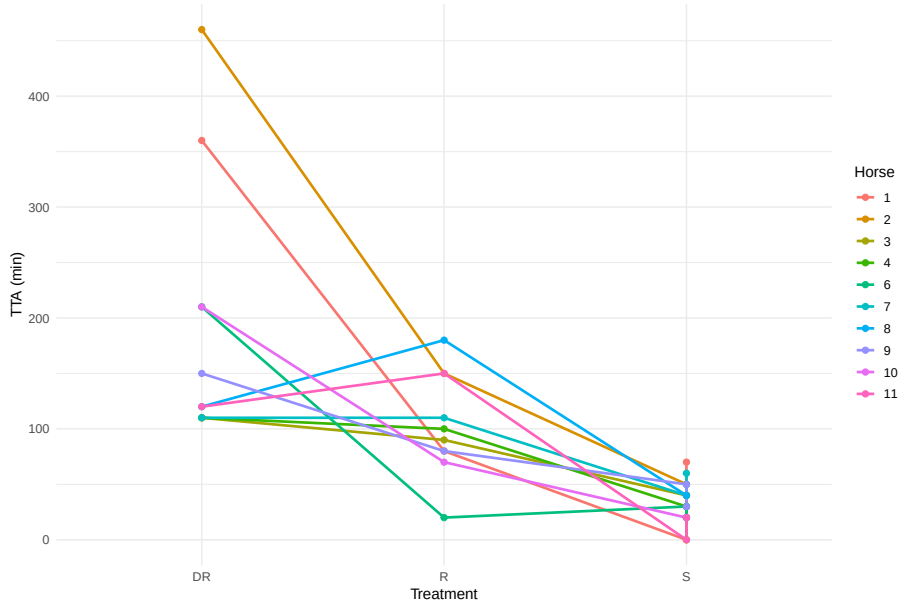
CTT was measured with a Cochet–Bonnet esthesiometer: a very fine, retractable nylon filament that is gently touched to the center of the cornea. The longer the filament, the less force it applies; the shorter it is, the more pressure it puts on the eye. If the horse blinks, the filament length is recorded in centimeters as the CTT. A CTT of 6 cm (maximum length) means the cornea is fully sensitive; a CTT of 0 cm (shortest length) means the horse felt nothing at any pressure: the cornea is maximally numb.

Study Design

- Ten healthy adult horses were enrolled (we excluded one eye due to cataract) in a **two-period crossover**. Each horse's two eyes took turns being the treatment eye vs. saline control, with assignments randomized and a one-week break in between.
- Before each session the horse was lightly sedated with xylazine and recorded baseline measurements:
 - **CTT** using the filament method above
 - **Corneal pressure** using a small tonometer
- After injecting the drug under the tissue around the cornea, these measurements were reported **every 10 minutes** for up to 2 hours (and then every 30 minutes if numbness persisted), until the CTT filament returned to its original length on two checks in a row.
- **Outcomes:** total anesthesia time (TTA), minimum CTT, duration of minimum CTT, maximum pressure; hemorrhage score (0–4); ambient humidity recorded.

Exploratory Data

Total Anesthesia Time by Treatment (Each Horse)



Analyzing Total Anesthesia Time

- **Why a mixed-effects model?**

- **Repeated measures:** each horse contributes two eyes $\times$ two periods
-> non-independence
- **Horse as random effect:** captures between-horse variability in baseline and response

- **Model specification**

- **Fixed effect:** Treatment (DR, R, S)
- **Random intercept:** Horse

- **Variance stabilization**

- **Square-root transform** on TTA to address right skew and heteroscedasticity

- **Inference**

- **Type III F-test** (Satterthwaite's approximation) for overall treatment effect
- **emmeans** on the Square-Root scale, back-transformed to original TTA (min) for interpretability
- **Paired Wilcoxon signed-rank test** (DR vs R) as nonparametric confirmation

Analyzing Total Anesthesia Time

Summary Statistics

Table 1: Summary Statistice for Total Anesthesia Time (Units: *mins*) Response

treatment	N	Mean	SD	Min	Median	Max
DR	10	196.0	121.5	110	135	460
R	10	103.0	46.7	20	95	180
S	20	31.5	19.5	0	30	70

Analyzing Total Anesthesia Time

Mixed Model (Square-Root-Transformed TTA)

Table 2: Type III F-test for sqrt(Total Anesthesia Time)

Term	Num DF	Den DF	F value	Pr(>F)
treatment	2	48	49.677	2.04e-12

Analyzing Total Anesthesia Time

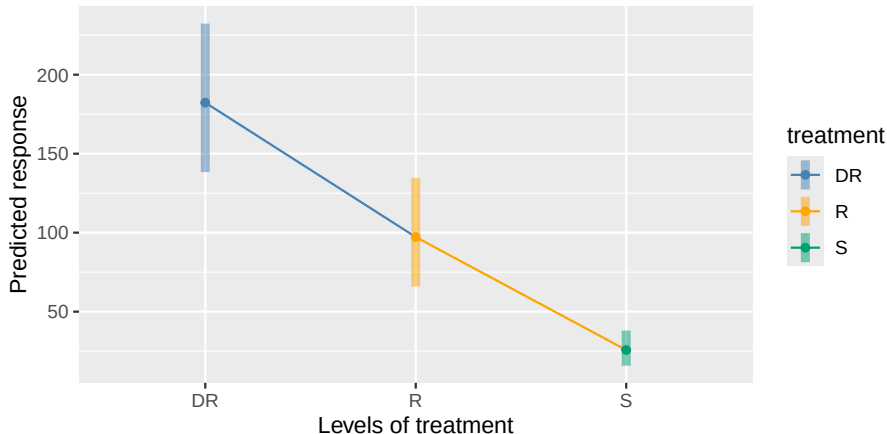
Estimated Marginal Means

Table 3: Tukey Pairwise Contrasts for Treatment (Sqrt Transformed Total Anesthesia Time (Units: *min*))

Contrast	Estimate	SE	DF	T-Ratio	P-Value
DR - R	3.641	1.14	48	3.19	0.00704
DR - S	8.432	0.90	48	9.33	4.05e-12
R - S	4.791	0.90	48	5.30	8.48e-06

Analyzing Total Anesthesia Time

Estimated Marginal Means



Conclusions

- Overall effect: $\sqrt{\text{TTA}}$ mixed model highly significant (Type III $F(2, 29) = 31.09, p = 5.9\text{e-}8$).
- DR vs R: DR increased $\sqrt{\text{TTA}}$ by 3.42 minutes ($\text{SE} = 1.26; t = 2.72; p = 0.028$). DR vs S: +8.43 minutes ($\text{SE} = 1.11; t = 7.58; p = 6.9\text{e-}8$).
- R vs S: R exceeded S by 5.01 minutes ($\text{SE} = 1.08; t = 4.63; p = 0.0002$).
- Model diagnostics show acceptable fit after square-root-transform and random effects.

Analyzing Total Anesthesia Time

Wilcoxon Test (DR vs R)

Table 4: Wilcoxon Test: Total Anesthesia Time (DR vs R)

Statistic	HL Estimate	95% Lower	95% Upper	p-value
38	105	-10	225	0.0756

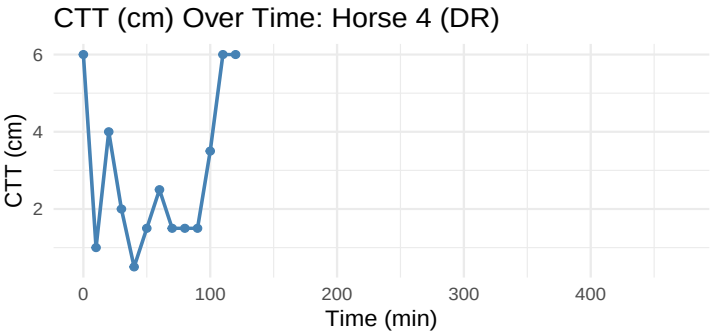
Conclusions

- Wilcoxon signed-rank (DR vs R), however, does not supports $DR > R$ ($p = 0.076$).
- *Note: The paired Wilcoxon signed-rank test for DR vs R (Statistic = 38; HL Estimate = 105; 95% CI = -102 to 225; $p = 0.0756$) was not significant. Because it only considers paired differences and ignores between-horse variability, its power is lower than the mixed model's, which uses all observations and models horse as a random effect.*

Full Time Course For A Single Horse

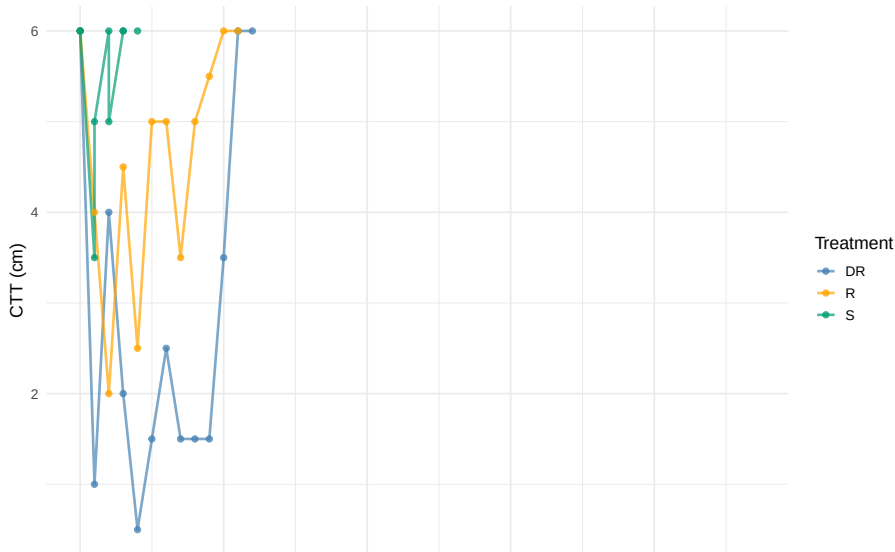
Table 5: Summarized Metrics for Horse 4 (Treatment DR)

Metric	Value
Minimum CTT (cm)	0.5
Duration of Min CTT (min)	10.0
Total Anesthesia Time (min)	110.0



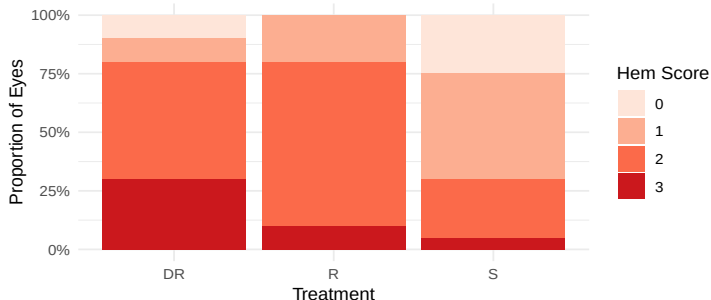
Full Time Course For A Single Horse (ALL Treatments)

CTT Over Time: Horse 4 Across Treatments



Exploratory Hemorrhage Data

Distribution of Hemorrhage Scores by Treatment



Wilcoxon Signed-Rank DR vs R test

Table 6: Wilcoxon Signed-Rank Test with Hodges–Lehmann CI

Statistic	H-L Estimate	95% Lower	95% Upper	p-value
12	0	0	1	0.8241